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DECLARATION PAGE

We declare that this report, which is based on our "GrowthMap" career pathway and course recommendation website, is our own work and has not been submitted previously for any academic qualification at this or any other institution. All sources of information used in this report have been properly acknowledged and referenced. We further declare that this work was carried out in compliance with the University's regulations on academic integrity and plagiarism.

ABSTRACT

Many students and working professionals find it difficult to select a career path or comprehend what qualifications they need in today's rapidly evolving job market. In countries like Sri Lanka, where there is a lack of clear accessible career guidance, this problem is even more difficult. People frequently invest in the wrong courses, acquire out-of-date skills or continue to be unemployed as a result. The Growth Map platform was developed to assist in resolving this issue. By taking into account their interests, objectives and present qualifications, this web based system assists users in finding appropriate career paths. Additionally, it exposes them to excellent and accessible educational opportunities.

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ABBREVIATIONS TABLE

Abbreviation	Full Form
AI	Artificial Intelligence
API	Application Programming Interface
CSS	Cascading Style Sheets
DB	Database
HTML	HyperText Markup Language
ICT	Information and Communication Technology
JSON	Javascript Object Notation
MVP	Minimum Viable Product
NVQ	National Vocational Qualification
OOAD	Object-Oriented Analysis and Design
RAM	Random Access Memory
WBS	Work Breakdown Structure
SQL	Structured Query Language
UI	User Interface
UX	User Experience

Chapter 1: Introduction

1.1 Chapter Overview

This chapter introduces the GrowthMap project, an AI-powered career planning and certification guidance platform. The system helps users identify suitable career paths according to their future aspirations and goals, assess their current skills, and find both high quality and affordable courses to achieve their professional goals.

The solution is then presented in the chapter, describing its in scope, out scope features and limitations.The chapter also describes the systems intended workflow and user journey, showing how each feature helps to provide a career-planning experience that is both accessible and meaningful.

The chapter also explains the motivation for creating the website and lays the foundation for understanding the motives,value and the direction of the system will be presented in the subsequent chapters.

1.2 Problem background

In today's fast changing job market, many students and professionals struggle to select a suitable career path and identify the right courses to achieve their goals. The rapid growth of technology, automation, and artificial intelligence has created new career fields while making some traditional jobs less relevant. However, students often receive limited guidance on which skills and qualifications are most valuable for their chosen paths.

One of the major challenges faced by learners, especially in developing countries like Sri Lanka, is the lack of proper career direction and affordable education options. Many young people are unaware of the qualifications required for specific careers or how to achieve the same level of knowledge through alternative, low cost programs. According to the Department of Census and statistics of Sri Lanka(2024), the youth unemployment rate among individuals aged 15-24 was around 25%, largely due to a mismatch between education and job market needs. This highlights a critical gap between academic learning and employability skills.

For example, a student aspiring to become a Software Engineer may believe that only an expensive degree from a top-ranked international university can secure a good job. However, many affordable online alternatives such as a Google's Software Engineering Certificate or Coursera's specialized programs offer nearly the same skill set at a fraction of the cost(Coursera,2025). Without proper guidance, students miss these opportunities and may end up spending unnecessary amounts of money or studying irrelevant courses.

Similarly, a person interested in becoming a Data Analyst might know which exact tools (like Excel, SQL, Python, and Power BI) are essential or which certifications are recognized globally. Without structured guidance, learners often follow random online tutorials without a clear learning roadmap, which leads to frustration and wasted time.

The problem becomes even more significant when considering global statistics. The World Economic Forum(2025) reports that 44% of workers' skills will be disrupted within the next five years, and over 85 million jobs are expected to be replaced by automation while 97 million new roles will emerge that require new skills. This means career planning and choosing the right courses have become essential for employability.

Therefore, there is a clear and urgent need for an intelligent platform that can guide individuals to:

- Identify the right career pathway based on their interests and abilities.
- Discover both premium and affordable course alternatives that deliver the same level of knowledge.

The proposed solution, an AI-powered career pathways and course recommendation website directly addresses the lack of a centralized career-guidance platform.This system ensures that users can achieve their dream careers without financial or informational barriers, making it user-friendly,locally relevant, and providing personalized career pathways supported by real-time job market insight , making the proposed solution both impactful and inclusive.

1.3 Problem Statement

Individuals find it difficult to find suitable career paths ,since information regarding necessary skills,certifications and industry requirements is dispersed, out of date, and unstructured.

1.4 Proposed Solution

The proposed solution is a Career Pathway & Course Recommendation Website designed to guide students, graduates, and professionals in Sri Lanka to make informed decisions about their career paths and learning opportunities. The platform aims to bridge the gap between the skills individuals currently possess and the skills required for their desired careers, offering a user-friendly and accessible digital solution.

The system provides a structured and personalized approach to career planning by identifying each user's strengths, interests and aspirations through an interactive profession-identification quiz conducted at the beginning of the user journey. Based on the results, the platform directs individuals toward the most suitable career pathways and learning directions aligned with modern industry demands.

The web application assesses current skills and identifies those that require improvement in order to guarantee that each user is aware of their place and potential within their selected professional path. The proposed solution analyzes the current qualifications the user possesses already, and proposes the required qualifications to achieve the user's goal. It offers easy-to-follow routes that enable people to confidently and effectively advance toward their intended positions. By offering reasonably priced educational options that are accessible both locally and online, the platform also guarantees accessibility and affordability.

By integrating relevant educational possibilities, educational institutions and employment data into a single, easily accessible platform, the solution enables users to easily investigate study paths that align with their objectives and unique situations. The platform is both locally inclusive and globally relevant, with support for Sinhala, Tamil, and English. Therefore, the suggested solution provides a dependable, user-friendly, and forward-thinking digital platform that enables people to create meaningful and well-aligned professional routes.

1.5 Aim

To develop a user friendly guidance and course recommendation platform that helps individuals identify suitable career pathways and identify both premium and affordable learning options based on their goals and preferences.

The aim of this project is to create a clear, structured, and easy to use platform that supports students and professionals in selecting the right career direction. Many learners struggle to understand what qualifications, skills, or courses are required for specific careers. This system will bridge that gap by providing career pathways, course recommendations, and learning roadmaps that match a user's interests, background, and aspirations.

The platform will also highlight affordable alternatives to expensive educational programs, giving users more choices regardless of financial limitations. By organizing career information in a simplified and accessible format, the system will help users make confident educational and professional decisions while reducing confusion, uncertainty, and unnecessary spending.

1.6 Project Scope

This section explains what will be included in the final product (in-scope) and what will not be implemented within this project timeframe (out-scope). The out-scope items are considered future enhancements that can be added later.

1.6.1 In-scope

1. Course Filtering Based on Personality and Future Career Aim

The system will allow users to filter available courses according to their interests, learning style, and future career goals. These filters help students narrow down their options so they can easily find courses that align with their preferred job field. This feature does not use AI; instead, it works with predefined categories and tags assigned to each course, ensuring simplicity while still supporting informed decision-making.

2. Budget Friendly Course Alternatives (Budget Filter + Comparison Engine)

Many learners cannot afford expensive programs, so the platform will display low-cost or free alternatives that offer similar knowledge. This comparison feature will show course prices, duration, delivery mode, and a basic overview. Users can quickly compare expensive courses with budget options to choose the most suitable and affordable learning path.

3. Geo-Mapped Learning Centers & Online Course Listings

The platform will include pages that display physical training centres and online course providers available in Sri Lanka. Although real-time map integration is out of scope, the system will show manually added location details, course names, brief descriptions, cost ranges, and contact information. This helps learners understand where relevant courses are offered without relying on advanced mapping technologies.

4. Real-Time Job Market Insight (Static/Manual Version)

The website will provide general job market information, such as trending fields, in-demand skills, and career opportunities. These insights will be manually researched and updated periodically. Since live job-market scraping and automated analytics require AI and APIs, the platform will only include static, summarized labour-market information.

5. Interactive Learning Progress (Simple Animated Interface)

A simple, visually appealing progress interface will allow users to track the number of courses they have completed or saved. Users can manually update their progress, and the dashboard will display animations or basic graphics. This feature motivates learners to stay consistent, even without AI-powered learning analytics.

6. Language Support (English, Sinhala)

The platform will include multi-language support to ensure accessibility for a wider audience in Sri Lanka. The main pages and key content will be available in all three languages so that students from rural and urban areas can use the website comfortably. This directly supports inclusivity and improves usability.

7. AI-Powered Personalized Roadmap

In the future, AI could analyze a user's skills, interests, and learning patterns to generate a customized career pathway. This roadmap would suggest the next steps, recommended courses, and skill-development timeline. However, such intelligence requires machine learning models and user profiling, which is outside the current project scope.

8. Skill Gap Analyzer (AI-dependent)

An advanced skill analyzer could evaluate a user's current knowledge and compare it with industry requirements for their chosen career. It would identify missing skills and automatically recommend what to learn next. Since this depends on AI assessments, it is out of scope for the current version.

9. Real-Time Job Market Insight (AI + API required)

Real time labour-market analysis would use AI tools and live job-portal APIs to track trending skills, job openings, and salary predictions automatically. Since this requires external data integration, it will only be included in later versions.

1.6.2 Out-scope

1. AI Career Mentor

A chatbot-style mentor capable of answering career-related questions, suggesting courses, and guiding the user throughout their learning journey requires AI algorithms and natural language processing. This makes it a future enhancement but not part of the initial build.

2. Career Outcome Predictor (AI Algorithm)

A predictive model that forecasts salary ranges, employability chances, and career growth after completing certain courses requires machine learning. This will be added in the future once user data and algorithms are available.

1.7 Rich Picture Diagram

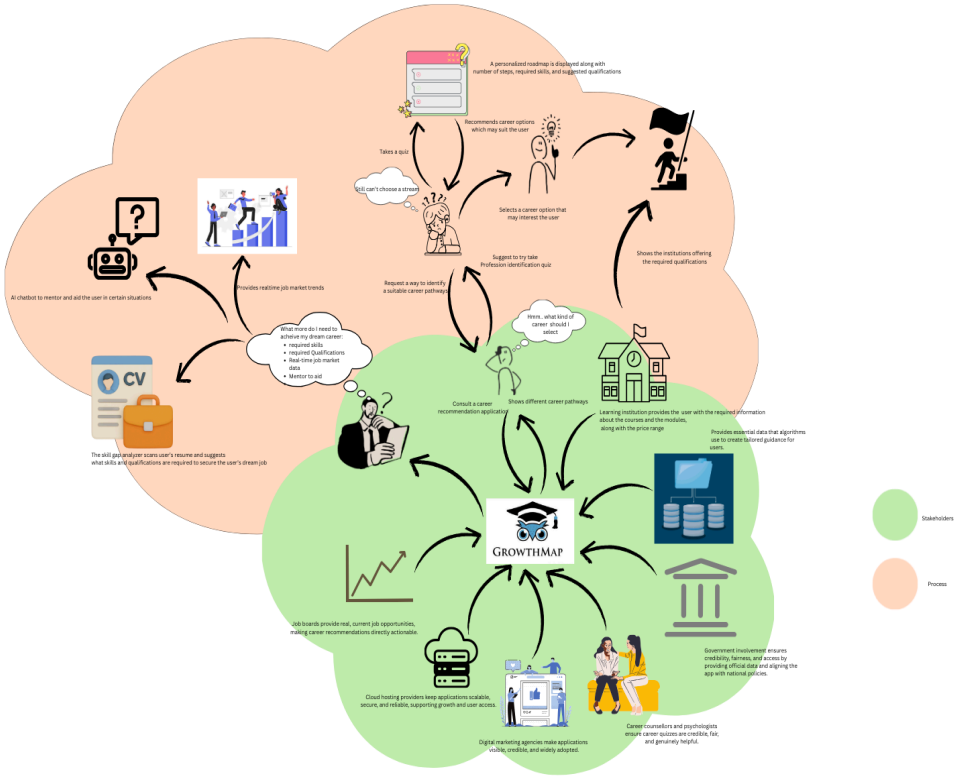

Figure 01 - Rich Picture Diagram

1.8 Resource requirements

Developing the minimum viable product(MVP) of the Career Pathway & Course Recommendation Website requires a combination of hardware, software, and relevant data. These resources ensure the system runs smoothly, *provides an* interactive experience, and supports the core features planned for MVP.

1.8.1 Hardware requirements

To develop and test the website effectively, a development computer or laptop with sufficient processing power and storage is needed. The system should have at least 8 GB

of RAM, a 256 GB SSD, and a modern processor such as an intel i5 or equivalent, which allows smooth operations of development tools and software. A stable internet connection is essential for accessing online courses, real time job market data, and map services. Additionally, testing the website across multiple devices including desktops, tablets, and mobile phones to ensure that the platform is responsive and accessible for all users.

1.8.2 Software requirements

The software requirements for the Career Pathway & Course Recommendation Website include both frontend and backend technologies, along with supporting tools for database management, visualization, authentication, version control, and hosting. The frontend will use HTML, CSS, and JavaScript to create the main interface, ensuring an intuitive user experience, while React.js can optionally enhance interactivity with dynamic, component-based elements.

The backend will be developed using Node.js with Express.js, managing server-side operations, routing, and API requests efficiently. A database such as MongoDB or Firebase will store user profiles, skill assessments, course details, and career recommendations. Chart.js will provide interactive dashboards for skill tracking and career insights, and Firebase Authentication will secure user accounts. Git and GitHub will handle version control for collaboration, while Firebase Hosting allows the MVP to be deployed and tested online. This combination of software tools ensures a functional, interactive, and secure platform for users to explore career pathways and learning opportunities.

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1.8.3 Data requirements (optional)

Although optional, having relevant datasets can greatly enhance the functionality, accuracy, and usability of the platform. This includes compiling comprehensive information on local and online course providers, covering details such as course fees, module contents, delivery formats, and location. Publicly available job market data, particularly from Sri Lanka, can provide valuable insights into trending skills and in-demand certifications, helping users focus on relevant learning paths. Furthermore,

maintaining a database of basic skill requirements for each career path allows the platform to perform skill gap analysis, guiding users to understand which areas they need to improve to achieve their career goals efficiently.

1.9 Business model canvas

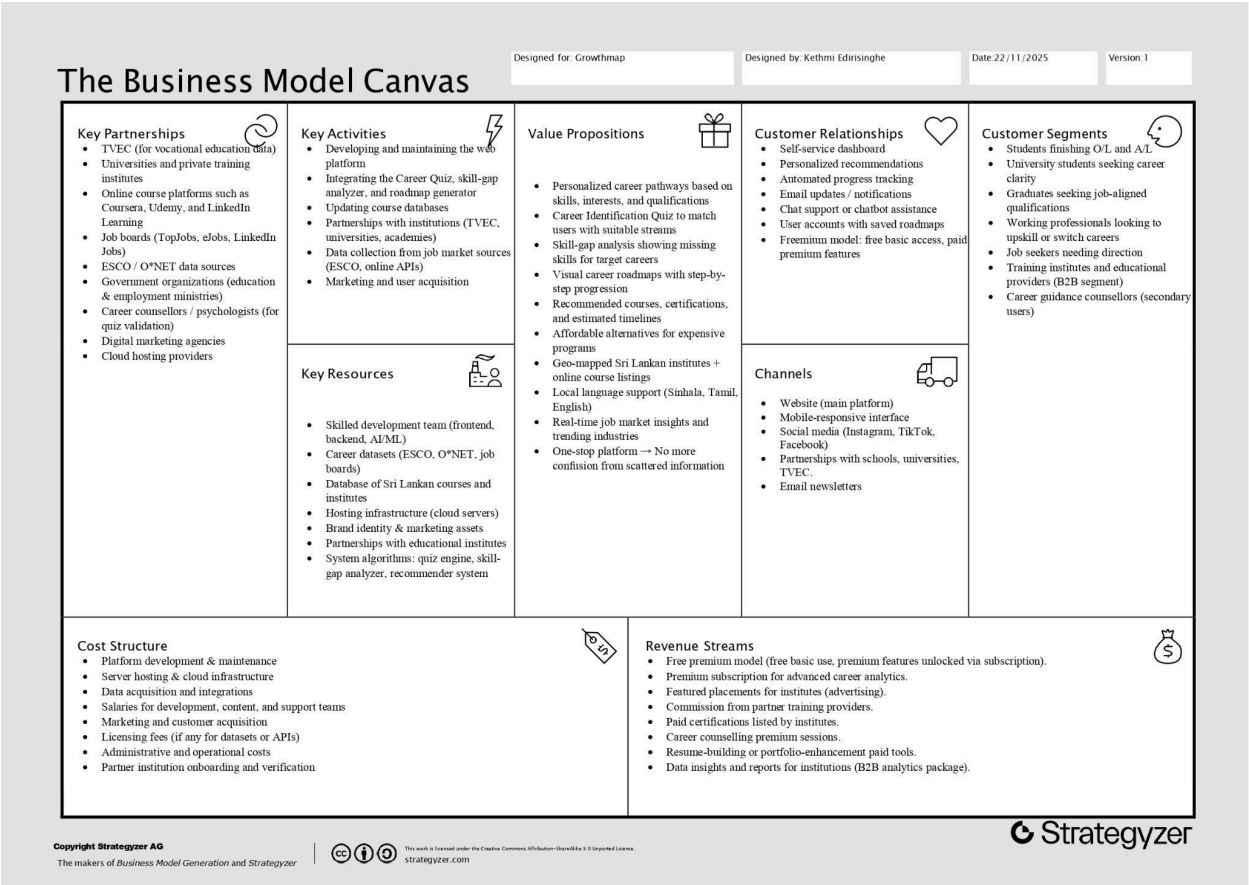

Figure 02 - Business Model Canvas

1.10 Chapter Summary

This chapter provided an overview of the Career Pathway & Course Recommendation Website project. It began by highlighting the problem background, emphasizing the challenges students and professionals face in choosing suitable career paths, identifying relevant courses, and

accessing affordable learning options in Sri Lanka. The problem statement succinctly captured the need for a platform to guide learners in making informed career decisions.

The chapter then presented the proposed solution, describing the development of a website that allows users to filter courses, analyze skill gaps, compare affordable alternatives, locate geo-mapped learning centers, and track learning progress through interactive dashboards. The project aim was clarified, along with the in-scope and out-of-scope features, distinguishing the minimum viable product from potential future AI-driven enhancements.

A rich picture diagram was suggested to visually represent users, features, and interactions, providing a simple overview of the system. The resource requirements were discussed, covering hardware, software, and optional datasets needed for development. Finally, the Business Model Canvas was introduced as a framework for understanding the operational, financial, and partnership aspects of the platform.

Overall, this chapter sets the foundation for the project by defining the problem, presenting a clear solution, outlining the scope, and identifying the resources and partnerships necessary to develop a practical, user-focused, and locally relevant career guidance platform.

Chapter 2: Existing work

2.1 Chapter Introduction

This is an overview of current technology and systems for predicting career paths and offering career advice is provided in this chapter. It compares the features and approaches of other platforms, looks at how they attempt to address similar problems, and identifies any gaps that our proposed system aims to address. Furthermore, our system's tools, procedures, and planned implementation approach are described as well.

2.2 Existing Work

As people look for more organized methods to plan their professional futures, career guidance platforms are growing in popularity. However, rather than offering a comprehensive career path planning experience, the majority of current systems only concentrate on a few specific areas, like career quizzes or job listings.

Existing Platforms andComparison Table.

Features / Platforms	CareerMe.lk	CareerOne.gov.lk	CareerExplorer.com	Teal.com	LinkedIn Career Explorer	Our Platform (ICT Career Guide)
Skill-based recommendations	✓	✓	✓	✓	✓	☑
Career path visualization	✗	✗	✓	✓	✓	☑ *(simplified, local-focused)*
Qualification roadmap (NVQ/Degree mapping)	✗	✗	✗	✗	✗	☑ (Unique Feature)
Local (Sri Lankan) relevance	✓	✓	✗	✗	✗	☑
Real-time dynamic recommendations	✗	✗	✓	✓	✓	☑
Integration with institutions	✗	✗	✗	✗	✗	☑ (Unique Feature)
Focus on academic + vocational mix	✗	✓	✗	✗	✗	☑
Free access	✓	✓	✗	✗	✓	☑
Personalized quizzes / user profile	✓	✓	✓	✓	✓	☑
Visual, interactive interface	✗	✗	✓	✓	✓	☑ *(student-friendly modern UI)*
Data-driven AI analysis	✗	✗	✓	✓	✓	☑ *(custom dataset for ICT)*

Table 01 - Existing Platforms and Comparison Table

Although the platforms examined in this report have some useful core features like basic assessment, career matching and other general guidance, they also have certain limitations in common. Many systems have the same weakness. There are no complete qualification roadmaps. Furthermore, they do not have NVQ levels, diplomas, degrees or any ICT related specific pathways for sri lankan students.. Tools such as careerme.lk or careerone.gov.lk are useful for their local relevance, but they rely on static databases and do not derive personalized dynamic pathways based on user's skill, goals or qualifications.

Many global career related platforms like careerexplorer.com, tealhq.com, LinkedIn Career Explorer offer latest analytics and global insight but they are not relevant to Sri Lanka, they aren't integrated with local institutes and they do not provide academic and vocational mapping suitable for Sri Lankan students, they mainly help people change jobs, but they don't tell students what education path they need to take.

However, our proposed solution seeks to fill these gaps with an ICT-related fields focused platform for our MVP.

- Skill based recommendations adapted for ICT fields.
- A qualification roadmap including NVQ Levels, diplomas, degrees, certificates and more.
- Connecting local, Sri Lankan institutions and recognized qualifications.
- Simple, student-friendly career visualizations

Moreover, a custom dataset trained for the Sri Lankan ICT domain possesses the capability to enhance the system.

Our platform now becomes much more relevant to local learners and much more comprehensive and personalized than any other platform.

2.3 Tools and Implementation plan

This system will be developed as a web-based career guidance platform that helps users to identify suitable career paths based on their interests and skills while suggesting relevant educational qualifications and local NVQ Levels or institutional options.

Our focus will be on building a Minimal Viable Product(MVP) with clear, functional workflow from user inputs to career recommendations.

Tools And Technologies.

Component	**Tool / Technology**	**Purpose**
FrontEnd	HTML, CSS, JavaScript	To design and structure the main user interface for user interaction
Frontend Framework	React.js	For creating dynamic, component-based user interfaces with better interactivity.
Backend	Node.js with Express.js	To handle server-side logic, routing, and API endpoints efficiently.
Database	MongoDB / Firebase	For storing user profiles, skill data, quiz results, and recommended career mappings.
Data Visualization	Chart.js	For displaying skill matches, career statistics, or qualification paths.
Authentication	Firebase	To manage user login, registration, and secure access.
Version Control	Git and Github	For code versioning and team

		collaboration.
Hosting	Firebase Hosting	To develop and test the MVP online

Table 02 - Tools and Technologies Table

Implementation Plan

Phase	**Objective**	**Task**	**Expected Outcome**
Phase 1: Requirement Analysis	Define the scope and gather relevant datasets	Identify skill categories, collect NVQ Levels and ICT qualification data from Sri Lankan institutions.	Structured dataset and finalized system scope.
Phase 2: System Design	Design both the frontend flow and backend architecture	Create flow diagrams (user flow, backend flow) and wireframes for the interface.	Clear Blueprint for development
Phase 3: Frontend Development	Develop the user interface	Build pages for registration, quiz/assessment, and career suggestions using HTML, CSS, and JS.	Functional, responsive UI prototype.
Phase 4: Backend Development	Implement the backend server.	Develop APIs to handle user data, store quiz responses, and fetch matching career data.	Working backend integrated with frontend.
Phase 5: Data Integration	Connect database with logic	Link skill-based questions with possible ICT career outcomes and corresponding qualifications.	Career recommendation system functional with real data.
Phase 6: Testing and MVP Launch	Ensure the system works as expected.	Perform functionality, usability, and data accuracy tests	MVP ready for demonstration and evaluation.

Table 03 - Implementation Plan Table

2.4 Chapter Summary

This chapter provides the foundation for the proposed solution by reviewing existing career guidance platforms and identifying their gaps in supporting Sri Lankan students, especially in ICT(Information Communication Technology) related qualification pathways. Throughout the comparison it became clear that current tools offer psychometric assessment, skill matching and job transition insights. But they lack localized qualification mapping, NVQ Level pathways and integrated institutes recommended which are essential for students navigating Sri Lanka's educational structure.

This chapter also introduced our proposed system concept of simplified but effective career guidance tool ICT domain as part of the MVP. The system aims to guide users from skill identification to relevant career options and then provide clear educational pathways that include local NVQ levels and available local institutions.

Overall, this chapter clarified the problem space and justified the scope reduction, and outlined a realistic and technically feasible plan for developing the MVP.

Chapter 3: Methodology

3.1. Chapter Overview

The Career Pathway & Course Recommendation System's general methodology is described in this chapter. It outlines the teamwork structure, time planning, collaborative tools, development methodology, design methodology, project management approach, and risk management techniques used during the project. The system is developed methodically, effectively, and in accordance with the project's goals thanks to the methodological framework.

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3.2 Development Methodology

The Agile Software Development Methodology was chosen for this project because it is flexible, iterative, and appropriate for developing a working Minimum Viable Product (MVP). As user needs and dataset structures become more apparent, the system's many evolving components such as career filtering, course recommendations, dashboards, and job-market insights need to be continuously improved.

Agile enables the development team to divide the system into manageable functional chunks, or sprints, with the goal of creating a usable feature at each sprint. Continuous improvement is made possible without waiting for the final deployment thanks to the feedback gathered during each iteration. For web-based applications, where UI/UX improvements and data integration frequently necessitate frequent adjustments, this flexibility is especially helpful.

Agile was chosen over traditional models like Waterfall because it supports:

- Progressive refinement of requirements
- Early testing and validation of components
- Continuous improvement during development
- Parallel development by multiple team members

- Faster delivery of core features for the MVP

Therefore, Agile provides an efficient, collaborative, and scalable approach for achieving the system's objectives.

3.3 Design methodology

The structure and behavior of the system were modeled using the Object-Oriented Analysis and Design (OOAD) methodology. OOAD is appropriate for a system with several interactive modules, including users, courses, careers, dashboards, and learning progress tracking, because it promotes modularity, reusability, and a clear separation of concerns.

The team was able to transform real-world concepts like "Course," "Career Path," "User," and "Skill Category" into well-defined objects with attributes and methods by using OOAD. To show system interactions and data flow, design diagrams like class diagrams, sequence diagrams, activity diagrams, and user interface mockups were made.

The selection of OOAD was motivated by:

- The modular nature of the system's components
- The ease of expanding the design to support future AI-enhanced features
- The ability to maintain consistent structure during integration
- Improved clarity for development and testing
- Enhanced maintainability and scalability

Overall, OOAD provided a robust design foundation for the system.

3.4 Project Management Methodology

The project team adopted the Scrum framework, a lightweight and structured approach under the Agile methodology. Scrum helps teams handle dynamic requirements, maintain accountability, and deliver consistent progress through time-boxed iterations.

Key Scrum practices used in this project include:

- Sprint Planning Meetings: Identifying tasks and assigning responsibilities for each sprint.
- Sprint Backlogs: Documenting upcoming tasks in priority order.
- Daily/Weekly Stand-Up Discussions: Brief team updates on progress, challenges, and next steps.
- Sprint Reviews: Evaluation of completed work and refining future goals.
- Sprint Retrospectives: Analyzing what went well and identifying opportunities for improvement.

Scrum was chosen because it supports collaboration, adaptability, and faster delivery of a functional MVP. It also ensures consistent communication among team members and provides transparency for all development tasks.

3.5 Team Work Breakdown Structure (WBS)

Work was divided among team members to ensure smooth progress and equal contribution. The Work Breakdown Structure (WBS) outlines how tasks were allocated.

Main Task Categories:

1. **Research & Requirement Analysis**
2. **Frontend Development**
3. **Backend Development**
4. **Database Structuring & Data Collection**
5. **UI/UX Designing**
6. **Testing & Quality Assurance**
7. **Documentation & Editing**

Example Contributions:

- **Member 1:** User interface design, frontend development (dashboard, progress tracker)
- **Member 2:** Backend logic, API creation, routing
- **Member 3:** Database schema design, course and career data aggregation
- **Member 4:** Researching job-market insights, NVQ levels, ICT pathways
- **Member 5:** Planning, documentation, formatting, preparation of diagrams
- **Member 6:** System testing and reporting issues

The WBS ensured clarity of responsibilities and avoided duplication of tasks, improving overall productivity.

3.6 Gantt chart diagram

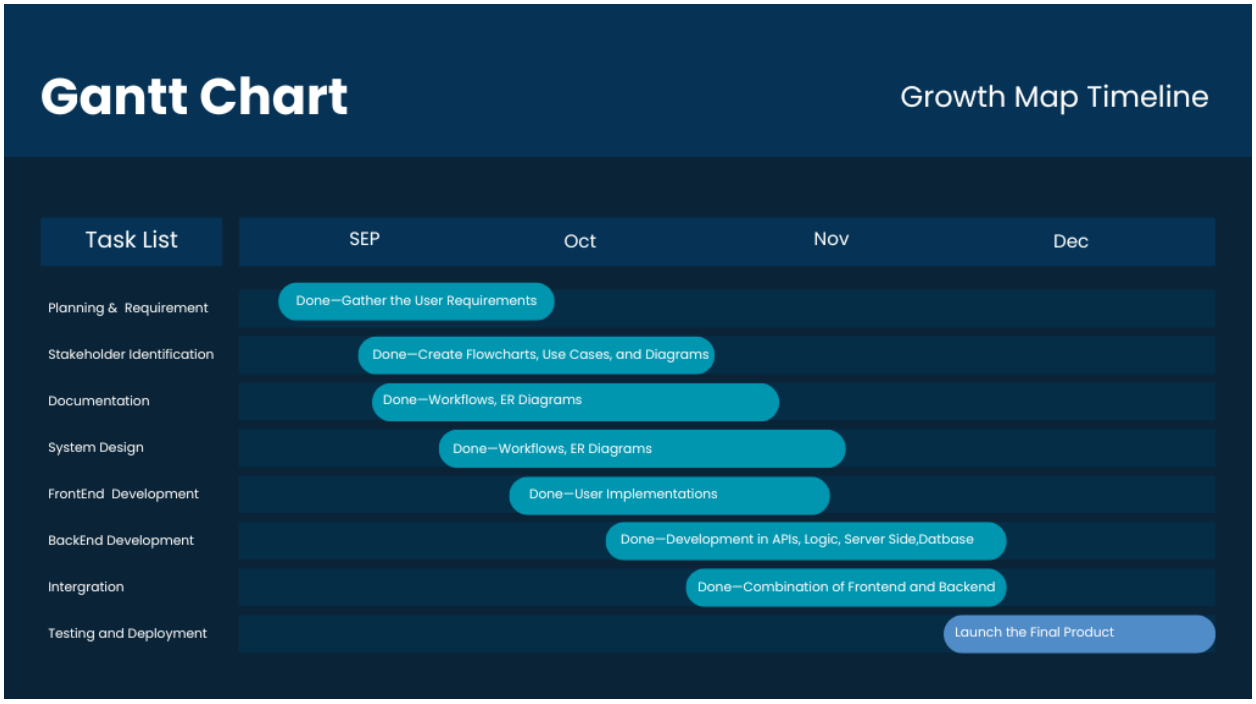

Figure 03 - Gantt Chart Diagram

3.7 Usage of Project Management and Collaboration Software In the project

The development of the GrowthMap web platform relied on a range of project management and collaboration tools to support coordinated teamwork and effective workflow management. These tools helped the team organise tasks, maintain clear communication, and ensure that project milestones were achieved on time.

Task Planning and Progress Tracking – ClickUp

ClickUp was adopted as the central platform for organising project activities and monitoring progress. The overall workload was divided into structured phases including Requirement Gathering, System Design, Implementation, Testing, and Final Documentation. Tasks were clearly assigned to individual members with defined deadlines and priority levels.

To support an Agile working approach, ClickUp was used to manage sprint activities and visualise task progress using status indicators such as Pending, Ongoing, and Completed. In addition, the platform was used to store shared resources including meeting summaries,

requirement notes, design screenshots, and progress updates, ensuring all team members had access to consistent and current information.

Source Code Management – GitHub

GitHub served as the primary platform for collaborative software development in the GrowthMap project. It enabled team members to work on features independently while maintaining a controlled and organised codebase. Pull requests and merge reviews were used to verify changes before integration, reducing errors and maintaining code quality.

Separate repositories were maintained for frontend and backend components, which improved clarity and simplified development, testing, and maintenance tasks.

Shared Documentation Workspace – Google Drive

Google Drive was used as a shared repository for all project-related documents, including reports, diagrams, UI mockups, and supporting evidence. The platform allowed multiple team members to edit files simultaneously, making collaboration more efficient and reducing version conflicts.

Team Coordination and Meetings – Google Meet and WhatsApp

Google Meet was used for scheduled discussions such as sprint reviews, requirement clarification sessions, and progress evaluations. These meetings ensured alignment across the team and provided opportunities to address challenges collaboratively. WhatsApp was used as a quick communication channel for updates, reminders, and informal discussions, allowing continuous interaction outside formal meetings.

3.8 Risks and Mitigation

Risk Item	**Severity**	**Frequency**	**Mitigation Plan**
Inconsistent or incomplete course data	High	Medium	Manual verification, periodic updates

Technical issues with hosting or deployment	High	Low	Backup hosting options and local offline testing
Miscommunication among team members	Medium	High	Weekly meetings and Trello updates
Dataset inaccuracies affecting recommendations	High	Medium	Cross-checking data sources and validating mappings

Table 04 - Risks and Mitigations Table

3.9 Chapter Summary

This chapter described the methodologies adopted for the development of the Career Pathway & Course Recommendation Website. Agile (Scrum) methodology was selected to ensure flexibility and iterative development, while OOAD was chosen to produce a structured and scalable design. Scrum-based project management supported task distribution, collaboration, and continuous progress tracking. The Work Breakdown Structure (WBS) clarified team responsibilities, and a detailed Gantt chart outlined the project timeline. Collaboration tools like Trello, GitHub, Google Docs, and Slack supported coordination throughout the project. Finally, potential risks were identified along with mitigation measures to ensure smooth project execution.

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Appendix

Each section under the appendix should be named "Appendix A", "Appendix B" etc.
For this the page numbering should be Uppercase roman numerals